

Thorne Aggregates

The model was built in forty minutes, presented to the board in twenty, and wrong by a factor of two, and every individual thing in it was the kind of thing a reasonable person does.

Thorne Aggregates was a crushed-stone and ready-mix concrete business operating eleven quarries and thirty-one plants across the Ohio Valley. Its founding family wanted to sell. Wrenfield Partners had four weeks to decide whether to bid, and the associate on the deal, Sam Okonjo, had built the valuation in the AI environment on the Thursday.

Okonjo's model produced an enterprise value of \$5,387 million and a value of \$74.81 per share. Wrenfield's investment committee had approved a first-round indication at \$70 on the strength of it.

The challenge came from Lisette Broussard, a partner not on the deal, in the Monday portfolio review. She had not read the model. She asked one question.

"Your terminal value implies you sell this business in year five at eleven and a half times EBITDA. Aggregates has traded between six and eight times for thirty years. What do you know that thirty years of transactions do not?"

Okonjo went back to the file. Over the next two days he found five things.

None of the five threw an error. None of them produced a number that looked absurd on the face of it. Each of them, taken alone, moved the valuation by between 5 and 24 percent, and four of the five moved it in the same direction.

The model's inputs were a starting revenue of \$1,480 million growing 7 percent a year for five years, an EBIT margin of 22 percent, depreciation of 6 percent of revenue, capital expenditure of 9 percent of revenue, an increase in working capital of 1.2 percent of revenue, a tax rate of 24 percent, terminal growth of 2.5 percent, and a weighted average cost of capital of 9.4 percent. Net debt was \$610 million and there were 72 million shares.

EXHIBIT 1 — WHAT OKONJO FOUND

- 1 The cost of capital in the model was 8.1 percent, not 9.4. The AI had been given the balance sheet and had computed the weights from book values of debt and equity. Thorne's equity was worth four times its book value.
 - 2 The terminal value was discounted for six periods rather than five. The loop that built the explicit forecast ran from 1 to 5, and the terminal value line used the loop counter after the loop had incremented past its last value.
 - 3 The tax rate appeared in an input cell as 24 percent. Inside the cash flow formula it was written as 0.21. Changing the input cell did nothing.
 - 4 Year 5 free cash flow included a \$46 million one-time gain from the sale of a depleted quarry parcel. The terminal value was built by growing year 5's cash flow at 2.5 percent forever.
 - 5 The change in working capital was entered with a positive sign in a formula that added it to cash flow.
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ASSIGNMENT

- 1 Build the corrected model and report the enterprise value and value per share. Compare with Okonjo's \$5,387 million and \$74.81.
- 2 Quantify each of the five defects separately: hold the other four correct and report the effect of each on enterprise value, in dollars and in percent.
- 3 Okonjo's per-share figure has a sixth defect that is not in the list of five. Find it, and rank it by dollar effect against the other five.
- 4 Broussard found all six from one number without opening the model. Name the check she ran, and construct three more checks of the same kind — outputs that can be tested against something known, without inspecting the calculation.
- 5 Rank the six defects by how likely each is to survive a careful review, and say what that ranking implies about where review effort should go.